HARE: errors that change size, errors that remember

A regression whose errors grow with a regressor, or carry over from one period to the next, or both: how to estimate it well, and how to tell which it is.

In Pollis	**Model** › **Linear Models**: *Linear Models with Heteroskedasticity*, *Linear Models with Autocorrelation* and *Linear Models with Heteroskedasticity + AR(1)*. Each panel's Code Example runs the package's estimators and tests on its data.
Missing in Julia	feasible GLS with a model of the variance (exponential, power, quadratic, linear, by group), by two-step or iterated weighted least squares and by maximum likelihood; the AR(1) estimators of Prais and Winsten, Cochrane and Orcutt, Hildreth and Lu, and Beach and MacKinnon; the two together; Harvey's and Glejser's tests. GLM.jl weighs by weights you give it, and robust standard errors leave the estimates as they are.
Package	`HARE.jl` (Heteroskedasticity and Autocorrelation Estimators), from GitHub: `Pkg.add(url = "https://github.com/Trumpingtons/HARE.jl")`.
Tests	1,335 tests of every estimator and test; 65 against R (prais, orcutt, skedastic, nlme, crch, lm) on the same data; 33 that recover the parameters from a million observations.

Why a package

Ordinary least squares stays unbiased when the errors are heteroskedastic or autocorrelated, but it is no longer the most precise estimator, and its standard errors are wrong. Robust (White, Newey–West) standard errors fix the second and leave the first. When the form of the variance or of the dependence is known, or can be estimated, generalized least squares fixes both. In R, these estimators are spread over prais, orcutt, nlme and skedastic; in Stata over `prais` and `hetregress`; in SAS most are in PROC AUTOREG. Julia had none of them, and Pollis' linear model panels needed all of them in one place, with the same interface, results that print as GLM.jl's do, and the estimate of the variance or of ρ beside the coefficients.

Where they are used

- **Cross sections whose spread grows with size.** Household spending varies more among the rich, firms' costs more among large firms, regional outcomes more where the population is small: the variance is a function of a regressor, the setting of Harvey (1976) and Glejser (1969).
- **Groups with variances of their own.** Countries, hospitals or survey waves measured with different precision: one variance a group (Greene 2018, §9.7.2).
- **Time series with persistent errors.** Demand or a macroeconomic relation fitted to annual or quarterly data, where a shock lasts beyond its period: AR(1) errors (Cochrane and Orcutt 1949; Prais and Winsten 1954).
- **Both at once.** A series whose shocks persist and whose volatility moves with a regressor; the estimators of the third panel, for which we found no other implementation.

The method

The model is $y_i = \mathbf{x}_i^\top \beta + u_i$, and the package estimates β with a model of u_i. Each function takes a formula and a table, `f(@formula(y ~ x1 + x2), data; Z = @formula(0 ~ x1))`, or matrices without a column of ones, `f(X, y; Z)`; $\mathbf{z}_i$, the regressors of the variance, are by default those of the mean.

Heteroskedasticity: a model of the variance

Link	Variance		Functions
Exponential	$\sigma_i^2 = \exp(\gamma_0 + \mathbf{z}_i^\top \gamma)$	Harvey	`exponential_fwls`, `_ifwls`, `_mle`
Power	$\log \sigma_i^2 = \gamma_0 + \sum_j \gamma_j \log \lvert z_{ij} \rvert$		`power_fwls`, `_ifwls`, `_mle`
Quadratic	$\sigma_i = \gamma_0 + \mathbf{z}_i^\top \gamma$	Glejser	`quadratic_fwls`, `_ifwls`, `_mle`
Linear	$\sigma_i^2 = \gamma_0 + \mathbf{z}_i^\top \gamma$	Breusch–Pagan	`linear_fwls`, `_ifwls`, `_mle`
By group	$\sigma_i^2 = \sigma_g^2$ for i in group g		`two_step_groupwise`, `iterated_groupwise`, `groupwise_mle`

Two-step FWLS fits OLS, regresses a function of the residuals on $\mathbf{z}_i$ (for the exponential link, $\log \hat{u}_i^2$, less $\mathrm{E} \log \chi_1^2 = -1.2704$, so that every γ is consistent and not only the slopes), and refits by weighted least squares with weights $1/\hat{\sigma}_i^2$. It needs no optimizer and stays consistent when the errors are not normal. *Iterated FWLS* repeats the last two steps until β stops moving, and for the exponential link converges to the maximum likelihood estimate (Oberhofer and Kmenta 1974). *MLE* maximizes the normal likelihood over β and γ together, by L-BFGS with ForwardDiff's gradients, started from the two-step

estimate, and gives standard errors for γ from the Hessian.

The FWLS standard errors of β are those of the weighted fit, conditional on the weights: $\hat{\sigma}^2(\mathbf{X}^\top\mathbf{W}\mathbf{X})^{-1}$ with $\hat{\sigma}^2 = \sum_i w_i\hat{e}_i^2/(n-k)$, as in R's `lm`, so they scale with the units of y. In the quadratic and linear links a fitted value can fall to zero or below, and no weight exists for such a row. FWLS and MLE then stop with an error that gives the number of rows and points to the exponential or power link, or OLS with robust standard errors. A value below 10^{-6} of the mean fitted value counts as zero.

Autocorrelation: AR(1) errors, $u_t = \rho u_{t-1} + e_t$

- **Prais–Winsten** (`two_step_prais_winsten`, `iterated_prais_winsten`): ρ from the residuals' first-order regression, then GLS on $y_t - \rho y_{t-1}$, keeping the first observation scaled by $\sqrt{1-\rho^2}$.
- **Cochrane–Orcutt** (`two_step_cochrane_orcutt`, `iterated_cochrane_orcutt`): the same, dropping the first observation.
- **Hildreth–Lu** (`hildreth_lu`): the ρ with the smallest sum of squares of the Cochrane–Orcutt transform, on a grid of `nrho = 200` points over $(-0.99, 0.99)$, about 0.01 apart; the global minimum on the grid, even when the sum of squares has more than one (Hildreth and Lu 1960).
- **Beach–MacKinnon** (`beach_mackinnon`): the exact likelihood, $\frac{1}{2}\log(1-\rho^2)$ included, concentrated in ρ and maximized by Brent's method (Beach and MacKinnon 1978).

Both: AR(1) errors with an exponential variance

$u_t = \rho u_{t-1} + \sigma_t e_t$, $\log\sigma_t^2 = \gamma_0 + \mathbf{z}_t^\top\gamma$. *Sequential* (`two_step_sequential`, `iterated_sequential`): Prais–Winsten first, then Harvey's regression on its residuals, then weighted least squares on the transformed data. *Joint* (`two_step_joint`, `iterated_joint`): the exact likelihood in ρ, γ and β together, with β concentrated out and $\rho = \tanh\xi$ to keep it inside $(-1, 1)$.

Tests

`harvey_test` and `glejser_test`: score tests of $\gamma = 0$ in the exponential and the quadratic link, χ^2 with as many degrees of freedom as $\mathbf{z}_i$ has columns, as R's skedastic computes them. `breusch_pagan_test` (Breusch and Pagan 1979), `white_test` (White 1980), `durbin_watson_test` and `breusch_godfrey_test(f, data, p)`, from HypothesisTests.jl, on the same formula; `wald_test` and `lrtest` between fitted models.

A worked example

The data

Five hundred periods with both problems: $y_t = 1 + 2x_{1t} - x_{2t} + u_t$, $u_t = 0.7\,u_{t-1} + \sigma_t e_t$, $\log\sigma_t^2 = x_{1t}$, the first u drawn from the stationary distribution, x_1, x_2 and e standard normal from `StableRNG(2026)`.

```
using HARE, GLM, StableRNGs
rng = StableRNG(2026); n = 500
x1 = randn(rng, n); x2 = randn(rng, n); sigma = exp.(0.5 .* x1); e = randn(rng, n)
u = zeros(n); u[1] = sigma[1] * e[1] / sqrt(1 - 0.7^2)
for t in 2:n; u[t] = 0.7 * u[t-1] + sigma[t] * e[t]; end
data = (y = 1 .+ 2 .* x1 .- x2 .+ u, x1 = x1, x2 = x2)
f = @formula(y ~ x1 + x2); Zf = @formula(0 ~ x1)
```

Which problem is it?

On the OLS residuals, `durbin_watson_test(f, data)` gives DW = 0.580 (residual autocorrelation 0.710, $p < 10^{-56}$) and `breusch_godfrey_test(f, data, 1)` $nR^2 = 253.4$. `harvey_test(f, data; Z = Zf)` gives LM = 11.47 on 1 degree of freedom, $p = 0.0007$: evidence of both, the autocorrelation much the stronger. The autocorrelation also weakens Harvey's test, whose residuals are not independent.

Four estimators

Estimator	x_1 (2)	s.e.	x_2 (-1)	ρ (0.7)	γ_1 (1)
OLS, `lm(f, data)`	2.178	0.088	−0.928	—	—
`iterated_prais_winsten`	2.084	0.048	−0.981	0.714	—
`exponential_mle(...; Z = Zf)`	2.105	0.083	−0.966	—	0.327
`iterated_sequential(...; Z = Zf)`	2.032	0.031	−0.985	0.715	0.937
`iterated_joint(...; Z = Zf)`	2.027	0.030	−0.985	0.704	1.016

Each correction for one problem alone helps; the estimators of both recover ρ and γ_1 and have standard errors of x_1 a third of OLS's and nearly two-thirds of Prais–Winsten's. The exponential MLE, which treats the errors as independent, finds a γ_1 of 0.327 against the true 1, and a standard error of the intercept of 0.085 against the joint estimator's 0.161. The joint estimator prints:

	Coef.	Std. Error	t	Pr(>|t|)	Lower 95%	Upper 95%
(Intercept)	1.17089	0.160694	7.29	<1e-11	0.855171	1.48662
x1	2.02746	0.029653	68.37	<1e-99	1.9692	2.08572
x2	−0.985216	0.0295266	−33.37	<1e-99	−1.04323	−0.927203
ρ	0.704261	0.0317495	22.18	<1e-99	0.642033	0.766489
log σ_0^2	0.0761484	0.0632498	1.20	0.2286	−0.047819	0.200116
z1	1.0159	0.0638266	15.92	<1e-56	0.890803	1.141

Log-likelihood: −266.5144 Converged: true Iterations: 7 N: 500 R^2: 0.5955

The sequential estimator's γ comes from residuals after Prais–Winsten, which reach it with ρ already estimated: its intercept, 0.33 against the joint 0.076, is not the variance's intercept of the model. Read γ from the joint estimator.

What the tests show

Against R

Five hundred observations from `StableRNG(42)` and `StableRNG(99)`, written to CSV and read by R 4.5.1; the values R printed are kept in `test/runtests_r.jl`, so the tests need no R. The differences are those measured on Julia 1.13.1 today.

HARE	R	Value	Largest difference
`iterated_prais_winsten`	prais 1.1.4	$\hat\rho$ = 0.747516	β 6.0e-10, ρ 6.7e-16
`two_step_prais_winsten`	prais, `twostep = TRUE`	$\hat\rho$ = 0.741971	β 5.4e-15, ρ 6.7e-16
`iterated_cochrane_orcutt`	orcutt 2.3	$\hat\rho$ = 0.747510	β 4.5e-10, ρ 5.6e-16
`beach_mackinnon`	nlme 3.1.168, `corAR1`, ML	$\hat\rho$ = 0.746041	β 1.7e-9, ρ 2.2e-8
`harvey_test`	skedastic 2.0.3	LM 122.0117; 1.1127	2.8e-13; 3.7e-13
`glejser_test`	skedastic 2.0.3	LM 188.1612; 0.8859	4.0e-13; 3.9e-13
`exponential_mle`	nlme, `varExp`	$\hat\gamma_1$ = 1.080067, ℓ = −694.6695	β 4.3e-11, γ 1.1e-9
`power_mle`	nlme, `varPower`	$\hat\gamma_1$ = 0.954741, ℓ = −565.8042	β 1.5e-9, γ 4.2e-8
`groupwise_mle`	nlme, `varIdent`	$\hat\sigma^2$ = 0.274, 0.758, 1.950, 3.171	β 4.2e-7, σ^2 5.8e-5
`quadratic_mle`	crch 1.2.3, identity link	$\hat\gamma_1$ = 0.504708, ℓ = −818.0891	β 1.2e-8, γ 4.6e-9

nlme maximizes the AR(1) likelihood by Newton–Raphson and HARE by Brent's method on the concentrated likelihood; they agree to 2×10^{-8} in ρ. nlme writes the exponential and power variances as $\sigma^2 e^{2\delta x}$ and $\sigma^2 |x|^{2\delta}$, so $\gamma_1 = 2\delta$. At its tightest setting nlme's group variances still miss the condition of the maximum, $\hat\sigma_g^2 = \mathrm{RSS}_g / n_g$, by up to 5×10^{-5} on data of 5,000 observations, where HARE's meet it to 10^{-15}.

Estimates and standard errors against R

The same data on both sides, each side with its own code: R 4.5.1, nlme 3.1.171, prais 1.2.0, orcutt 2.3, crch 1.2.3, Julia 1.13.1. Each table gives x_1 in full and, for the three coefficients, the largest difference. First the estimators R can run from start to finish.

HARE	R	x_1 HARE	x_1 R	s.e. HARE	s.e. R	Largest diff.
`exponential_mle`	`gls, varExp`	1.993399	1.993399	0.031057	0.031127	β 5.8e-11, s.e. 1.3e-4
`power_mle`	`gls, varPower`	2.115026	2.115026	0.048315	0.048390	β 1.5e-9, s.e. 7.5e-5
`groupwise_mle`	`gls, varIdent`	1.974048	1.974048	0.036187	0.036296	β 4.2e-7, s.e. 1.1e-4
`quadratic_mle`	crch, identity link	2.004756	2.004756	0.069204	0.069204	β 1.2e-8, s.e. 2.1e-9
`beach_mackinnon`	`gls, corAR1`	2.000847	2.000847	0.032530	0.032530	β 1.7e-9, s.e. 1.4e-8
`iterated_prais_winsten`	prais	2.000888	2.000888	0.032507	0.032507	β 6.0e-10, s.e. 5.2e-9
`two_step_prais_winsten`	prais	2.000733	2.000733	0.032595	0.032595	β 5.0e-15, s.e. 1.0e-15
`iterated_cochrane_orcutt`	orcutt	2.000953	2.000953	0.032542	0.032542	β 4.5e-10, s.e. 4.7e-9

The three likelihood estimators give standard errors a little below nlme's. For `groupwise_mle` the ratio is 0.99700, which is $\sqrt{(n-k)/n}$ and consistent with a degrees-of-freedom correction in nlme. For the exponential and power links, where HARE's come from the joint Hessian, the ratios are 0.99777 and 0.99846.

We found no R function for FWLS as a whole, so the next table compares its weighted fit: R's `lm` receives the same data and HARE's weights, and, for the sequential estimators, HARE's transformed data. It checks the fit and its standard errors, not the estimate of the weights. The quadratic and linear rows use $Z = 1 + |x_1|$, because on these data $Z = x_1$ gives fitted values below zero and both stop with the error above.

HARE, against `lm(...; weights = w)`	x_1 HARE	x_1 R	s.e. HARE	s.e. R	Largest diff.
`exponential_fwls`	1.993232	1.993232	0.030998	0.030998	β 5.9e-15, s.e. 5.6e-17
`exponential_ifwls`	1.993378	1.993378	0.031110	0.031110	β 1.7e-11, s.e. 8.0e-12
`power_fwls`	1.947179	1.947179	0.058686	0.058686	β 2.2e-15, s.e. 1.4e-17
`power_ifwls`	1.947097	1.947097	0.058670	0.058670	β 2.3e-10, s.e. 4.6e-11
`quadratic_fwls`	1.989685	1.989685	0.065363	0.065363	β 1.3e-15, s.e. 2.1e-17
`quadratic_ifwls`	1.989642	1.989642	0.065354	0.065354	β 2.7e-15, s.e. 2.1e-17
`linear_fwls`	2.017984	2.017984	0.077288	0.077288	β 1.3e-15, s.e. 2.8e-17
`linear_ifwls`	2.021275	2.021275	0.078790	0.078790	β 6.1e-11, s.e. 1.3e-11
`two_step_groupwise`	1.972670	1.972670	0.036595	0.036595	β 1.6e-14, s.e. 2.6e-16
`iterated_groupwise`	1.974048	1.974048	0.036296	0.036296	β 1.2e-10, s.e. 2.0e-11
`two_step_sequential`	1.998674	1.998674	0.032391	0.032391	β 3.0e-15, s.e. 5.6e-17
`iterated_sequential`	1.998436	1.998436	0.032267	0.032267	β 5.9e-13, s.e. 3.3e-12

Large samples

With a million observations from `StableRNG(123)`, 33 tests check that the two-step FWLS of each link, the MLEs, the four AR(1) estimators, the groupwise estimators and the sequential and joint estimators recover β, γ, ρ and the group variances within 0.01 (`test/runtests_consistency.jl`, 88 seconds).

The rest

`test/runtests.jl`, 1,335 tests in 32 sets, on 200 observations with known parameters: each estimator by formula and by matrix, two-step against iterated, the coefficient names, `predict` on new data, the Wald and likelihood-ratio tests, t or normal inference, standard errors that scale with y, and the error when a fitted value reaches zero.

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Package written and tested with Claude Code (Anthropic) in Pollis.

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